Nixor O Levels
Test Week 1
Additional Mathematics
Grade 10

Student Nam_____ ___ Time: 60 minutes

Student ID: _____ Max marks: 35

Section : _____

Please read and follow the instructions below:

- Please write your name, section and ID on the question paper.
- Do not turn over the paper unless instructed to do so.
- Only write in blue or black ink.
- Use of correction fluid is strictly prohibited.
- If you have any questions, raise your hand to ask the invigilator.
- Omission of working will result in loss of marks.
- Use a pencil for making straight lines.
- Answer all questions.
- Calculators not allowed.

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35)

List of formulas

Equation of a circle with centre (a, b) and radius r.	$(x-a)^2 + (y-b)^2 = r^2$
Curved surface area, A, of cone of radius r, sloping edge l.	$A = \pi rl$
Surface area, A, of sphere of radius r.	$A = 4\pi r^2$
Volume, V, of pyramid or cone, base area A, height h.	$V = \frac{1}{3}Ah$
Volume, V, of sphere of radius r.	$V = \frac{4}{3}\pi r^3$
Quadratic equation	For the equation $ax^2 + bx + c = 0$, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Binomial theorem	$(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \ldots + \binom{n}{r}a^{n-r}b^r + \ldots + b^n$, where n is a positive integer and $\binom{n}{r} = \frac{n!}{(n-r)!r!}$
Arithmetic series	$u_n = a + (n-1)d$ $S_n = \frac{1}{2}n(a+l) = \frac{1}{2}n\{2a + (n-1)d\}$
Geometric series	$u_n = ar^{n-1}$ $S_n = \frac{a(1-r^n)}{1-r}$ $(r \neq 1)$ $S_\infty = \frac{a}{1-r}$ $(\lvert r \rvert < 1)$
Identities	$\sin^2 A + \cos^2 A = 1$ $\sec^2 A = 1 + \tan^2 A$ $\operatorname{cosec}^2 A = 1 + \cot^2 A$
Formulas for ΔABC	$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ $a^2 = b^2 + c^2 - 2bc\cos A$ $\Delta = \frac{1}{2}ab\sin C$

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PCCja

1 (a) Express $2x^2 - 8x + 3$ in the form $a(x+b)^2 + c$, where a, b and c are constants. [3]

~~$2(x^2 - 8x$~~ $2(x^2 - 4x + 2^2 - 2^2) + 3$

$2[(x-2)^2 - 4] + 3$

$2(x-2)^2 - 8 + 3$

$2(x-2)^2 - 5$

$2(x^2 - 4x + 2^2 - 2^2) + 3$

~~$2(x^2$~~

$2(x-2)^2 - 5$

3

$x \quad -8$
3
-24

(b) Hence write down the coordinates of the minimum point of the curve $y = 2x^2 - 8x + 3$. [1]

$2, -5$

$2(x-2)^2 - 5$

$-(-2), -5$

$(2, -5)$

1

DCCja

(c) On the axes below, sketch the curve $y = 2x^2 - 8x + 3$, stating the coordinates of the minimum point and of the point where the curve crosses the y-axis. [3]

DCCja

y int $= 0, 3$

TP $= 2, -5$

$x = 0 \Rightarrow y = 3$

$x = 1 \Rightarrow$

$2x^2 - 8x + 3$

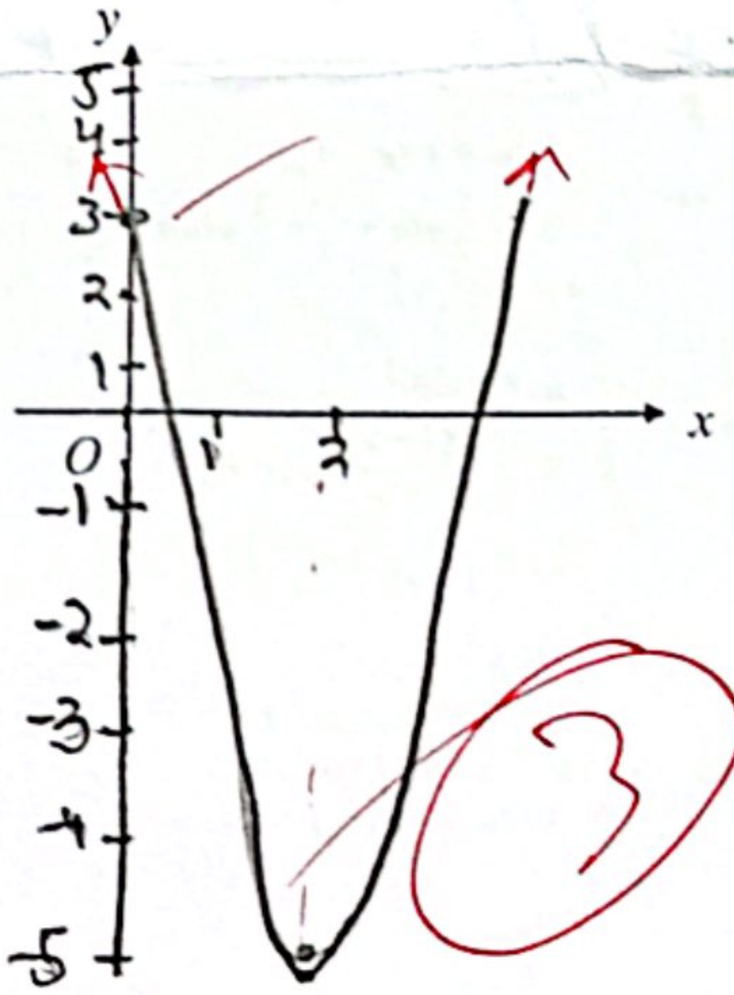
$\frac{8 \pm \sqrt{64 - 4 \times 2 \times 3}}{4}$

$\frac{8 \pm \sqrt{64 - 24}}{4}$

$\frac{8 \pm \sqrt{40}}{4}$

$2.5 \leftarrow \begin{array}{r} 1.25 \\ 1.25 \\ \hline 6.25 \\ 12.50\,x \\ 1\,25\,x\,y \\ \hline 1.5\,6\,25 \end{array}$

2.25



3

$2x^2 - 8x + 3$

$2[x^2 - 4x + 2^2 - 2^2] + 3 =$

$+2 = \cup$

$2(x-2)^2 - 5 = 0$

$2(x-2)^2 = 5$

$(x-2)^2 = 5/2$

$x - 2 = \pm\sqrt{5/2}$

$x = 2 \pm \sqrt{5/2}$

$y = 2 \times 0 - 8 \times 0 + 3$

$y = 0 - 0 + 3$

$y = 3$

8

$64 - 4 \times 2 \times 3$

$= 40$

8

$64 - 4 \times 2 \times 3$

DCC jq

2 Solve the inequality $2x^2 - 5x - 3 \leqslant 0$. [3]

$2x^2 - 5x - 3 \leq 0$

$2x^2 - 6x + x - 3 \leq 0$

$2x(x-3) + 1(x-3) \leq 0$

$(2x+1)(x-3) \leq 0$

$2x+1=0 \quad x-3 \leq 0$

$x = \frac{1}{2} \quad x = 3$

$x = \frac{1}{2} \quad x = 3$

$-\frac{1}{2} \leq x \leq 3$

3

$b^2 - 4ac \leq 0$

+	−	+
$x < -\frac{1}{2}$	$-\frac{1}{2} < x < 3$	$x > 3$

$3x-2-5x2-3$
$-4+10-3$
-73

$3x0-5x0-3$
$0-0-3$
$=-3$

$2x4^2-5x4-3$
$2x16-20-3$
$32-23=9+$

3 Find the remainder when $4x^3 + 8x^2 - 3x + 5$ is divided by $2x - 1$. [3]

DCC jq

$2x-1=0$

$2x=1$

$x=\frac{1}{2}$

$4(\frac{1}{2})^3 + 8(\frac{1}{2})^3 - 3(\frac{1}{2}) + 5$

$4 \times \frac{1}{8} + 8 \times \frac{1}{4} - \frac{3}{2} + 5$

$1/2 + 2 - 3/2 + 5$

$\frac{1}{2} - \frac{3}{2} \Rightarrow -1 + 7$

$R = 6$

3

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Paper

4 The polynomial p is given by $p(x) = 6x^3 - x^2 - 5x + 2$.

Write $p(x)$ as a product of linear factors.

$p(x) = 6x^3 - x^2 - 5x + 2$

$6(-1)^3 - (-1)^2 - 5(-1) + 2$

$6(-1) - (1) + 5 + 2$

$-6 - 1 + 7$

$-7 + 7 = 0$

$0 = 0$ ✓

$x = -1$

$x + 1 = 0$

$(x+1)$

$6x^3 - x^2 - 5x + 2$

$6, -1, -5, 2$ |4|

-1	6	-1	-5	2
	↓	-6	$+7$	-2
	6	-7	2	0

$6x^2 - 7x + 2 = 0$

$6x^2 - 3x - 4x + 2 = 0$

$3x(2x-1) - 2(2x-1) = 0$

$(x+1)(3x-2)(2x-1)$

4

5 The line $y = 3x + c$ is a tangent to the curve $y = x^2 + 5x + 7$.

(a) Find the value of c. [4]

$3x + c = x^2 + 5x + 7$

$c = x^2 + 5x - 3x + 7$

$c = x^2 + 2x + 7$

$c = x^2 + 2x + 1^2 - 1^2 + 7$

$c = (x+1)^2 - 1 + 7$

$c = (x+1)^2 + 6$

$c = 6$

4

Perfect

$b^2 - 4ac = 0$

$-7 +$

$3x + c = x^2 + 5x + 7$

$0 = x^2 + 2x + 7 - c$

$0 =$

$2^2 - 4 \times 1 \times (7 - c)$

$4 - 28 + 4c = 0$

$4c - 24 = 0$ [2]

$4c = 24$

$c = 6$

Perfect

$x = \frac{-(5)}{2(1)}$

(b) Hence find the coordinates of the point of contact.

$3x + 6 = x^2 + 5x + 7$

$0 = x^2 + 2x + 1$

$0 = x^2 + x + x + 1$

$0 = x(x+1) + 1(x+1)$

$0 = (x+1)(x+1)$

$0 = (x+1)^2$

$x = -1$

$y = 3(-1) + 6$

$y = -3 + 6$

$y = 3$

coords = $(-1, 3)$

2

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PCCIA $\frac{14\frac{1}{2}}{9}$ / 15

6 Solve the equation $2x^3+5x^2-4x-3=0$. [5]

$2x^3+5x^2-4x-3=0$

$2(1)+5(1)-4(1)-3=0$

$7-7=0$

$0=0$

$x=1$

$(x-1)$

1	2	5	-4	-3
	↓	2	7	3
	2	7	3	0

$2x^2+7x+3=0$

$2x^2+6x+x+3=0$

$2x(x+3)+1(x+3)=0$

$(x-1)(2x+1)(x+3)=0$

$x=1 \quad x=-\frac{1}{2} \quad x=-3$

5

DCCIA

7 The polynomial $f(x)=x^3+ax^2+bx-6$ has a factor of $x-1$ and leaves a remainder of -12 when divided by $x+2$.

$f(1)=0$

$x=1$

$1+a+b-6=0$

$a+b-5=0$

$a+b=5$

$a=5-b$

$a=5-3$

$a=2$

(a) Find the value of a and of b. [4]

~~$1+a+b-6$~~

$1^3+a(1)^2+b(1)-6$

$1+a+b-6=0$

$a+b-5=0$

$a+b=5$

$a=5-b$

$a=5-3$

$a=2$

$(-2)^3+a(-2)^2+b(-2)-6=-12$

$-8+4a-2b-6=-12$

$4a-2b-14=-12$

$4a-2b=2$

$4(5-b)-2b=2$

$20-4b-2b=2$

$-6b=-18$

$b=3$

$a=2$

4

DCCIA

(b) Using your values of a and b, show that the equation $f(x)=0$ has exactly one real root. [3]

x^3+2x^2+3x-6

1	1	2	3	-6
	↓	1	3	6
	1	3	6	0

$(x-1)\ x^2+3x+6=0$

$(x-1)\ \frac{-3\pm\sqrt{9-4\times1\times6}}{2} \rightarrow \frac{-3\pm\sqrt{9-24}}{2} \rightarrow \frac{-3\pm\sqrt{-15}}{2}$

$x=1$

$3^2-4\times1\times6$

$9-24 \rightarrow -15$

3

The discriminant is -15 which is less than zero hence roots for x^2+3x+6 do not exist meaning $x=1$ coming from $(x-1)$ is the only real root.